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Managing strategic-level human-machine teams in the agribusiness sector

Santiago Ismael, PAVON

Under the direction of: **Mr Olivier, DUPOUËT**

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Abstract

As we are entering the doorsteps of the fourth industrial revolution, the emergence of disruptive new technologies is the new-normal to our daily lives. However, the consequences and shifts they provoke to our daily and future lives is a matter of importance. Particularly over this writing, we will venture into the near future possibility of having to share the workplace with machines and saying hello to hybrid human-machine teams at organizations. We are not thinking about harvesting machines or autonomous aspirators, but rather independent and complex robots and software that could work side by side to middle and top managers across several sectors. But as we food production is a key element of human life and industrial production, let's just start by projecting this scenario into the agribusiness sector. Which elements and which constrictions could we find in the future concerning this hybrid human-machine teams at a medium or higher organizational level ?

Key words: HRI (Human Robot Interaction), human-machine teams, Artificial intelligence, co-evolution approach, interface & interaction experience, agribusiness sector.

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Introduction

Stepping into the year 2023 we are entering fully into the fourth industrial revolution. A revolution that will be characterized, in part, by the widespread use of robots and robotics in every work environment and in all industries. But particularly, the agribusiness sector will be one of the first ones to be changed by this disruptive agent: the robot. Let it be by the labour-intensive work needed for this industry or the inherent attractivity that this sector has in incorporating new technology.

However, the reality will go beyond this point, and passing the operating level tasks that are currently being shared by humans and robots, it will come a time when managers would need to work alongside machines in their day-to-day routines. This is a turning point not only for society in general but for the managerial sciences, never have we ever faced such a radical shift in the way management works and will work. Moreover, how can we even direct and govern this hybrid machine-human teams where each agent (human and robot) will have its own set of characteristics, motivations, limiters, ways of interacting, etcetera.

As professor Henny Admoni (Admoni, 2020) said while speaking about the future of human-robot interaction in a Davos 2020 conference[1], we should let people do what they are good at and let robots do what they are good at. In a way we are acknowledging that we, humans, have our limitations and robots theirs, so a common path would be to each one focus on their capacities that outperform the other. Having this in mind, we can ask what the limitations on the human side will be for working in a managerial level alongside machines.

More specially, focusing on the agribusiness sector, so we can have a clear description and study of the topic. What plausible constraints could arise within the agribusiness sector in the near future that would make managing a human-machine team different from a regular human-human team? And more interestingly, not only the differences but also which conflicts could arise, to whom to listen and to trust first, how to resolve these differences. As seen, these questions are of highly importance for the upcoming years managers and their relationship with their colleagues, whereas human or machines.

[1] https://www.youtube.com/watch?v=2uRQXFF9VI8&list=WL&index=115

Narrowing down our path within this topic, the research problem can be defined as: What plausible constraints could arise within the agribusiness sector in the near future that would make managing a human-machine team different from a regular human-human team? Cutting it by part, we can identify key aspects of the research problem. There are plausible constraints on managing a human-machine team; we are locating in the near future (+2 to +5 years ahead); and the differences from a regular human-human team to a human-machine team.

Focusing on the first key aspect, we can come up with a series of research question as followed:

- Which conflicts could arise when managing a human-machine team at a managerial level?
- Which differences can be pointed from a human-machine team from a human-human team?
- Could be trust issues when a machine argument goes against a human argument?
- Is there a specific set of characteristics for a human-machine team that will work on the agribusiness sector?

The aim of these set of questions will be to give a clearer path on what to expect from a human-machine team? what set of conflicts and their characteristics can we find? And specifically, to the agribusiness sector, would there be individual characteristics on a human-machine team for this sector and/or constraints? Converging we are trying to find out more about what the future of working at a managerial level on human-machine teams will be like. For approaching on having an answer for these questions, a literature review involving three main topics was involved. Starting with a global view of Ai and ethics, human robot interaction and ways of studying relations between humans and machine and finally, machine behaviour.

After the writing of the literature review, two main concepts came as basic for the conception of any team in general, but even more importantly to the realization of hybrid human-machine teams : trust and communication. For this matter, two hypotheses were thought, to trying to give a clearer insight on the weight of these two values over the establishment of prosperous hybrid human-machine teams and which approaches or positions in practice could help future managers to promote this two-core value in the future. The hypotheses are:

1. *Trust within a human-machine team is higher when the members employ a co-evolution approach rather than an individualistic one.*

2. *Communication in human-machine teams within the agribusiness sector will be higher when using robots with a human-like interface and overall interaction experience rather than a computer-like one.*

For the methodology, a quantitative approach will be implemented. Obtaining data using a questionnaire that was responded from over 7 countries and with 40 respondents varying from age group, level of education and field of work. The respondents answered using a 1 to 5 scale on whether they totally disagreed or not over a specific set of questions and phrases, these phrases were specially created to be used as indicators for the study of the variables over the hypotheses. For hypothesis 1, which approach would be more correct to promote trust, and for hypothesis 2, which interface & interaction experience would promote communication. The questionnaire was done online and at times where the novelty of Ai is at an increase due to several news concerning new generative Ai software's launched to the public in recent months.

The results obtained affirm that a co-evolution approach is of better interest to enhance trust over hybrid human-machine teams, and also at the agribusiness sector specifically (although at a lower level). On the contrary, there was a negative response that having a human-like interface & interaction experience to promote communication was the right approach. Nevertheless, an interesting position could be pointed as for the answers from people working over the agribusiness sector, this negativity was highly reduced. Further studies could be made to better and more profoundly study the differences on why and whether this differences between the responses from people working at the agribusiness sector are more pro human-like interface or not from the general responses.

The outline will follow a literature review over three main concepts and topics: Ai, Human Robot Interaction and Machine Behaviour. Then a discussion on the hypotheses creatin and their relevance, methodology of the study and questionnaire used, data description, data analysis and finally a conclusion and further plausible studies to have in the future.

Literature Review:

Introduction

Reminding the research question addressed: What plausible constraints could arise within the agribusiness sector in the near future that would make managing a human-machine team different from a regular human-human team? As the issue discussed is a novel topic, the focus on the academic research will be on trying to find recent articles and studies were possible (from 2017 and on) as to have the latest developments and investigations on the subject.

The interest of the discussion in the fields related to Ai throughout the scientific community is palpable and visible. From Stanford University's One Hundred Year Study on Ai (AI100)[2], and to the words of Demis Hassabis (CEO of Google DeepMind parallel company focused on Ai): "First solve Ai, then use Ai to solve everything else". Following (Glikson & Woolley, Human Trust in Artificial Intelligence: Review of Empirical Research, 2020) "future leaders should be able to manage machine-machine, human-human, and human-machine teams". Hence, we can clearly state that Ai, human-machine teams, and the future scenarios at the workplace that the mass integration of these technologies will mean for workers, managerial sciences, and managers themselves is an important and contemporary question of the times that will be over a few years ahead.

In the next literature review part, we will focus on deepening our knowledge on four key points that will help us better understand the globality of the situation where we are standing on and it's intricacies.

We will start with the broad concept of Ai, as a central subject for our discussion and the base for all the posterior concepts developed. Here, we will review several points concerning Ai and it's conception: intelligence, algorithms, human subjectivity, trust, upskilling, value alignment problem and Ai ethics.

Moving forward to human-robot interaction, a new field of studies that tries to understand the complex dynamics that develop between humans and Ai enhanced machines (robots). Here we will introduce why studying the relation that humans and robots have is of central interest to our investigation although we also consider Ai-powered software and not only robots, for this reason, key concept ideas developers use to characterize the relation of these hybrid teams. Also, the specificities of a human-

[2] Review the annual reports made by the university with the latest news about Ai at: *aiindex.org.*

machine relation over the agro-industrial sector, plausible consequences for organizations on mass employment of robots and lastly, some possible scenarios and issues regarding HRM strategies to cope with hybrid teams on a managerial level.

And finally, machine behaviour; to help us grasp the intricacy of the characteristics of machines actions and study plausible constraints that would reflect on human-machine teams.

Key words: artificial intelligence, human-robot interaction, agriculture robotics, machine behaviour.

Artificial intelligence

Although the first concepts and discussions about artificial intelligence and intelligent machines could go back centuries, we will focus from one turning point in history; and that would be Alan‘s Turing discussions about computing machinery and intelligence. The so-called *Turing test* was envisioned as a thought experiment where a human interrogator would ask written questions to rather a computer or human. If the interrogator couldn't differentiate if the answers given were from a human or computer then the answer to: Can a machine think? would be answered (Turing, 1950).

Starting from this pivotal moment and study, we will use the massive +1.100 pages long reference book of Russel & Norvig, Artificial Intelligence: A modern approach (2021) as an initial guide on our quest to grasp the profundity of the topic. However, a first introductory concept shall be explained, the agent. An agent (*agere* from Latin, to do) is simply put something that acts (Russell & Norvig, 2021), and moreover, something that acts rationally or with a subsequent set of logic behind it, a set of goals and motivations, a rational agent.

Continuing with the authors study of Ai, they start with intelligence itself in several aspects but questioning that intelligence could be defined in terms of the human performance of intelligence: a comparison to the human mind intelligence (Russell & Norvig, 2021). But another approach could be made as to see the acts behind the agent and to see how similar to human actions that would be (this is the approach Turing had in its test). Also, the idea of an agent thinking rationally, and from here even Aristotle with his syllogisms over patterns of argument structures, could be used as an example of how an agent can think logically. And lastly, an agent that acts rationally, to study their innate intelligence throughout their actions instead of the thoughts processes.

To help us grasp how an artificial agent could act and think, we can start from the concept of algorithm. The word itself derived from Muhammad ibn Musa al-Khwarizmi, a 9th century mathematician, it means: "a set of mathematical instructions or rules that, especially if given to a computer, will help to calculate an answer to a problem" (Cambridge University Press, (n.d.)). Following (Hill, 2015) thought of algorithms, as a "procedure or decision process" concerning a mathematical construct "accomplishing a giving purpose under given provisions". Algorithms are a tool for computers to provide an answer/action to a set of pre-established thinking processes within a specific environment.

But the "nature" of algorithms and Ai enhanced agents themselves is not free from the influence of human subjectivity (Mittelstadt, Allo, Taddeo, Wachter, & Floridi, 2016). The subjectivities of the humans while the design and configuration processes are passed into the artificial agents. Then the decision-making processes of these agents could be biased, as clearly (Macnish , 2012) put it: "the values of the author, wittingly or not, are frozen into the code, effectively institutionalising those values''. Considering these biases must be present when studying how an Ai agent works and thinks, the decision taken reflects the human societal biases and subjectivisms; specially when considering human-machine teams where robots would work alongside humans.

Trust in Ai has deep importance over the topic (Hengstler, Enkel, & Duelli, 2016), as a common factor for researchers, communication is a key factor for promoting the acceptability of innovations between humans. The algorithms and "thinking" processes of the artificial agent must be understandable and communicated to the human users/co-workers of the team in order to gain trust. Following of also the purpose that the agent has throughout its actions. Finally, the study suggests that the gradual introduction of new technologies is a must in order to enhance the acceptance and trust of the innovations proposed (as several other authors also see the high levels of initial resistance and risk perceived by humans) (Heidenreich & Spieth, 2013).

Trust can be also deterred and impacted by the history made of preceding analogous scenarios (previous industrial revolutions and new technologies at the time and their percussion on societal views on technology). For this reason, "comparing previous

waves of automation and robotization" (Ernst, Merola, & Samaan, 2019) can enlighten our base knowledge of the subject.

Starting from the point that new technologies and advancements in society always come up with an equal negative force fearing the advancements themselves. Shifts in productivity can be expected, as also new skills can become desirable as opposed to others that become rapidly obsolete. As in the case of application and adaptation to new technologies, those at the forefront of research and implementation of Ai could benefit from an easier starting point. However, there have been already issues with human workers resistance over the fact that AI-powered robots could make them lose their jobs and truly be a teamwork but more about an automation (were the machine takes over the task) instead of augmentation paradigm (where human capabilities are expanded/magnified by the interaction with robot-Ai). (Ivanov & Webster, 2019) (Smids, Nyholm, & Berkers, 2020). Moreover, this comes as a psychological issue over the mass adoption of emerging technologies, we can call this term "computer or technology anxiety" (Ahmad, Cary, Zaheer, Ismail, & Imran, 2021), this anxiety could manifest into specific states of "frustration, apprehension, fear and uneasiness" (Okumus, Bilgihan, Ozturk, & Zhao, 2017). All these psychological factors and fears could deteriorate and block the future acceptance of human-machine operations in the workplace and the new realities that having this hybrid teams could mean for their daily lives.

Specifically in our case, Ai is a general-purpose technology, meaning it can deeply and rapidly change how we work on a vast span of sectors however the skills level of the person involved or the industry. Counting that the emergence of improvements and their scale in the field of Ai are becoming clearer each year, and the changes they make more profound and disruptive. Ai has an enormous potential for shifting the labour market in several ways; from the cost-advantage of labour seen in low-income countries by the possible emergence of mass market robots (automation and re-shoring due to robotization of tasks) (Ernst, Merola, & Samaan, 2019), a vast re-education process needed by the middle classes in order to keep up with the new technologies (and avoid deskilling), or unthinkable productivity gains for almost all possible tasks and process already in place by humans. Specifically for agriculture, if farmers can get better information, faster and deeper data management and more assertive resolutions on how to produce their crops, the potential is enormous.

A topic in itself, the upskilling required in society to fully embrace the benefits of Ai, shouldn't been left out of the conversation. As discussed by (Ernst, Merola, & Samaan, 2019), the skills and tasks that could remain in human control and hands are the one 's requiring "strong social, empathic, and interpersonal competences" and not exclusively relying only on the technical skills. Meaning that social and emotional skills will be central for a correct upskilling of the workforce as technical skills could be more easily and rapidly being taken over by artificial agents. Education and implementation will be once again (compared to past industrial revolutions) a massive factor for countries and societies to take advantage of Ai instead of living its negative effects. As discussed before, the flux of new technologies and work paradigms could make existing skills obsolete and the reasoning for building up new competencies that could become meaningless overnight (Henfridsson, Mathiassen, & Svahn, 2014) makes it hard (or it will be hard) to keep up the pace.

Connected to trust, we can discuss the "value alignment problem" as seen in (Sutrop, 2020). The author takes the idea from (Wallach & Allen, 2009) that as robots take more responsibility, they need to be programmed with moral decision-making abilities. These are needed for the sake of their well-functioning over a human environment where moral decisions are being taken at every aspect of work and at all times, even when we aren't aware. There has to be a matching of the human and machine objectives, to pursue objectives and goals that are of human interest and benefit.

A central author for this issue is Russell's idea that to maintain a control over the artificial agent, the Ai's should be "humble, altruistic, and committed to pursue our objectives, not theirs" (Russell S. , 2019). In order to achieve these goals, he proposed three guidelines that could help developers and scientists in their pursuit. The first one discusses that machines solely objective should be to maximise the realization of human preferences. Secondly, as these preferences are unclear and unsettled for the artificial agent. And finally, the most prominent source of data about human preferences and values is human behaviour in itself. Summing up all three guidelines, intelligent artificial agents should infer the human values throughout observation, simulation or even, impersonation. Adding a final touch to the discussion, Russell clarifies that he would prefer using human preferences over human values as it takes out the moral depth and discussion out of the dilemma and gives the opportunity to developers to focus on the behavioural aspect of human condition more clearly.

As we, humans, are not thoroughly rational and perfectly logic agents, machines need to take this consideration in hand to emulate our behaviour properly. As they will be envisioned by humans, on our personal evaluations of what we prefer the agents to do on a specific situation.

Moving to an ethical point of view of artificial agents, (Sutrop, 2020) proposes the idea that machines should be able to "distinguish between moral and immoral decisions" and empowered to take morally correct decisions when faced the situation.

Amitai Etzioni and Oren Etzioni (2018) discuss that for the machine to be a moral agent with autonomous decision-making capacities, it should have a set of attributes and characteristics. The first one being self-conscious, the agent needs to be aware of itself and its surroundings and of all possible courses of action that he may take in the situation. Secondly, he needs to be aware of the fact that he can alter that reality. Thirdly, the agent should be able to the moral principles and guidelines to be used when a particular moral choice takes place. And finally, the agent should be integrated with a motivation on acting morally, such as a will, intention or passion being programmed on his existence beyond just intellectuals' preferences of choice.

This path of argumentation leads us to the idea that ethics should be aligned with Ai, instead of just aligning Ai with values and preferences. For this reason, the programming of Ai should be highly interdisciplinary, and not exclusively for computer scientist, as the actions and concepts of artificial agents are much beyond just their technical field. There is an imminent danger that if we don't consider the programming of artificial agents on this multidisciplinary approach and also being involved in the ethics integration into Ai, there could be a simplification or dogmatism of Ai ethics.

Annette Zimmermann and Bendert Zevenbergen (2020) discuss seven traps within Ai ethics discussion: reductionism, oversimplification, the relativism trap, value-alignment trap, dichotomy; the myopia trap; and the trap of the rule of law. On overall, the authors conclude that Ai ethics should be open to value pluralism instead of a dogmatic value relativism with one answer only applicable to all situations and backgrounds. Nevertheless, keeping in mind that we could always list paths of actions that are comparatively better than others. Reinforcing this position, is the (European Commission, 2019) view that "AI systems should respect the plurality of values and choices of individuals", moreover, Ai should also be aligned with "the values of respect for human rights, democracy, and rule of law". Considering a publication from the IEEE

Global Initiative on Ethics of Autonomous and Intelligent Systems (IEEE, 2019), it mentions that "If machines engage in human communities as autonomous agents, then those agents will be expected to follow the community's social and moral norms. A necessary step in enabling machines to do so is to identify these norms. But whose norms?".

For Ai to remain beneficial for humans it should be aligned with human values, however, this could lead to a very strict and short list of human preferences. Considering ethics and morality as a whole could be of much help for the discussion. A helping ground (Sutrop, 2020) could be seeking agreement on the negative values, the one's we don't want artificial agents to perform. As we have discussed about pluralism over the good values, moralities, and visions, we have an easier task if we start by seeking agreement on what if morally wrong; for all of us.

Human-robot interaction (HRI)

Although our discussion focuses over human-machine teams, meaning artificial intelligent agents in form of powerful software embodied over a machine or directly a robot, we will see that discussing HRI is potentially useful; the why's will be discussed at the end. Human-robot interaction (HRI) is a fairly new field of study, from computer science to a much more specific focus on the relations between mechanic artificial agents (robots) and humans. HRI more specifically is "multidisciplinary research area that investigates ways to understand, design, and evaluate robotic systems and their interactions with humans" (Vasconez, Kantor, & Auat Cheein, 2019). The collaboration between the two agents is a crucial point of study, and all their dynamics, differences and plausible conflicts are of interest for this field. Here, we can study the physical, cognitive, and social synergies present on their relation. And from this relation, we can focus on "designing, understanding, and evaluating the interaction between humans and robots that can communicate and/or share physical space" (Nomura, Billard, Burke, & Murphy, 2010). Some examples of different applications of the study of HRI on specific interactions could be on humanoids, vehicle navigation, rescue robots, assistant robots, agricultural robots, drones, etcetera.

Following (Vasconez, Kantor, & Auat Cheein, 2019) idea on key concepts designers use and develop to face the challenges in different HRI scenarios, we can take a deeper look into them to characterize what makes HRI a specific field of study.

The first one being Level of Autonomy (LOA), describing the degree of automation of a robot, and from an HRI point of view, at which level the interaction could the machine interact with humans and on what degree of autonomy this interaction can happen. There could be three plausible scenarios according to Vasconez et al, direct control, where the user‘s decision is required for acting, although reducing the total work overload. A dynamic autonomy side, where the focus is on having natural, fluid, and efficient interaction with the human agent, and finally, a mixed-initiative strategy, were whether the human and/or robot intervene appropriately to the situation and action in place.

Secondly, the nature of information exchange, describing the channels through which the robot can communicate, and the protocols followed to doing so. Thirdly, the structure of the team, identifying the composition of the work teams and the number of respective members (humans and robots) and the roles each one takes within the team. Fourthly, adaptation, from the interaction the humans and robots face, how can they adapt, train, learn and even evolve from the experience. And finally, task-shaping, technically of the desired task is done, how it should be done and if the HRI focus can modify any of it‘s processes.

As our focus of discussion is specifically in the agribusiness sector, HRI strategies are of key help in order to provide solutions to “complex problems, providing security, comfort, lower work- load, and better process productivity” (Vasconez, Kantor, & Auat Cheein, 2019).

Specifically for agriculture, artificial agents will need to be able to learn and adapt to different working conditions and use human‘s long history and knowledge of the agribusiness sector. Currently, HRI architectures on the sector are focused on manual and production task; however, we can already see a growth on more complex software that focuses on the planification of work, analysis of productive field data and action planning, logistics and economics of industrial processes, etcetera. This task represents highly complex processes that require plenty of years for a human to study and to master but some software’s are already in place to equal the humans’ capacities in some specific tasks as discussed above.

Coming back to our initial point on why HRI is useful for a discussion although we talk about human-machine teams (Ai in the form of software counts too) and not specifically about robots, HRI gives a potent insight at the ways we can approach the

communication between the two agents. We already have plenty of experiences, data, and analysis over the relation in teams of humans and robots, what are the general issues that arrive at work, how can we promote trust or more interestingly, what causes the distrust from the human side while working with a robot. What are the designs, ways of communicating and "expressing" that could be more human-friendly and receptable, how do people approach working with an artificial agent, what do they expect, how do they collaborate with them, if they even do. All these questions can be easily extrapolated to a general case on human-artificial agents' teams, were trust, good communication, safety, fluidity, and receptiveness will show vital for its success.

Now we will discuss the above interrogations and check what do past experiences and HRI tells us. A major double point to start with is safety and trust. As discusses by (Cheein, y otros, 2015), "safety should be a key design consideration […], especially if humans and robots work cooperatively sharing space" and no mystery that as physical agents that could move and be independent, robots could easily trigger concerns over the security of their nature and actions.

Also (Benos, Bechar, & Bochtis, 2020) consider the study of safety and ergonomics during human-robot interaction in the agricultural sector as an emerging field. They also present five scenarios where robots in agriculture could work (consider for this case for operative work, nonetheless a valuable addition to the discussion). The first one being in isolation, here the purpose is avoiding undesired contact, hence the robot will stay on a specific area of work, direct physical HRI is not possible in this scenario. Another case could be in coexistence, where the barriers are lifted, although the robot has the capabilities to detect and avoid the human presence-touch in its working zone. Thirdly is synchronization, where the tasks one agent does are quickly followed by the other, allowing a much visible interconnection between both. Next is cooperation, robot and human operate within the same area at the same time, although they have different task. And lastly, collaboration, were both agents execute their work together and in a synergetic manner with the help of all sensors and systems put together. These scenarios and their advantages can be studied at previous analysis made by (Berenstein & Edan, 2017) (Adamides, y otros, 2017) (Bergerman, y otros, 2015)

Now we can study how employment and organizations might have changed due to robot implementation, a work done by (Dixon, Hong, & Wu, 2020). Firstly, they discuss how the adoption of robots could reduce monitoring and control jobs as robots

are repetitive, independent agents most of the time (besides agency issues, robots do not have this problem as they do not work for self-interested objectives such as humans do); and this supervision labour is generally made by managers at the organizations. Indirectly meaning that the job composition might start to change already due to this factor; although also an increase in high skills jobs (software engineers, developers, computer scientist, etcetera) that could sustain these artificial agents are another factor to reckon with. A new business re-engineering might be on the way, were both low-skilled jobs (operative) and highly skilled jobs (managerial and especially middle management) could be started being taken over by machines.

As robots are measured by their productivity, fastness, efficacity and efficiency, an increase on performance pay on individual performance by humans could be predicted according to the authors. Although, the authors also found that plenty of tasks robots couldn‘t do because of the nature of the activities, these activities requiring tacit knowledge. This knowledge being hardly codified and programmable into data or instructions robots could consume (Marcus & Davis, 2019).

And now let‘s get closer to our question about human-machine teams at a strategical level (and not operational as largely studied before), there some authors who have come close to these questions as well. (Ahmad, Cary, Zaheer, Ismail, & Imran, 2021) introduces the idea that having machine-human interaction in the workplace will bring a “plethora” of issues and challenges to HRM managers (and not only to HRM managers in our discussion) , and also how will managers work and be measured for their work (Richards, Yeoh, Chong, & Popovic, 2019). Consequently, Ai systems and tools (robots) will be intertwined with Human Resources Management (HRM) and make unprecedent changes on how the managerial sciences will be affected by this new wave of challenges.

Some authors (Ahmad, Cary, Zaheer, Ismail, & Imran, 2021) reflect on the idea that human workers will expect machines/Ai to do every task perfectly, consequently, having a reduced communication at the workplace. (Schaefer, 2013). On the contrary, another scenario is when humans start to intervene in the tasks managed by the artificial agent, fearing that they won‘t be done as efficiently if not (Demir, McNeese, & Cooke, 2020). This behaviour can also be affected whether the agent is a robot, or an Ai powered software (Glikson & Woolley, 2020).

Finally, Ahmad et al, 2021, give us 4 major hurdles and potential HRM concerns over the interaction of humans and Ai over four different issues (not always specifically

to robots but altogether is a concrete study of the relationships between humans and artificial agents).

1. “ Workers’ anxiety of working in the same team with AI-powered machines/robots” (Smids, Nyholm, & Berkers, 2020).
    a. Possible hurdles: Resistance by some workers in accepting these artificial agents as new team members.
    b. Possible HRM strategies: Having a clear communication protocol before the introduction of these hybrid teams. Having wide training initiatives for the human workers and ensuring a adequate level of technological competence.
2. “Trust between human workers and Ai” (Ivanov & Webster, 2019) (Glikson & Woolley, 2020) .
    a. Possible hurdles: Fear of losing jobs. An inability to properly communicate with an Ai-powered teammate.
    b. Possible HRM strategies: Having a facilitating environment, clearly stating the purpose and objective of hybrid-teams and sustaining a communication flow with human workers about their situation and future developments.
3. “Tasks expectations from each other” (Schaefer, 2013) (Demir, McNeese, & Cooke, 2020).
    a. Possible hurdles: Human workers under or over-estimating Ai-powered team members capabilities.
    b. Possible HRM strategies: Clear task specification, expected results of the system and a clear view of the flow of tasks.
4. “Performance evaluation of team members” (Backlund, Maurin Söderholm, Engström, Andersson Hagiwara, & Lebram, 2018) (Kim, Kim, Kim, & Dey, 2018).
    a. Possible hurdles: Human workers have natural limitations (need to eat, sleep, take holidays or become ill) whether artificial agents do not and their performance doesn‘t fluctuate as human (Jacob W. Crandall, y otros, 2018)s could.

b. Possible HRM strategies: Contextualising the performance of the human agents by adding their natural limitations on the performance evaluation frameworks on a team and organizational level. [3]

The authors (Ahmad, Cary, Zaheer, Ismail, & Imran, 2021) summarize that there will be major psychological barriers and fears over HRI at the workplace, although they propose several counteractions in order to avoid or to flatten the consequences/reactions of the human team members; they emphasize that we don't have enough information or guidelines concerning what type of leadership styles that would be advantageous for hybrid human-machine teams. They also emphasize the plausible study of human reactions and development when working on hybrid teams concerning their emotions in place and their adjustments, the worker's resilience, and adaptability, and what will be the main psychological barrier and who needs to be eliminated first for humans when working with artificial agents. And lastly and most importantly, what will be the collective response of impeding ("resistance to change, fear of job loss, organisational inertia") and driving forces ("training, conducive environment, collaboration, communication, performance improvement") on the interaction between hybrid human-machine teams.

Over this section we have introduced the concept of HRI, giving us a depth of the relation that machines and humans have (and could have), their intricacies, difficulties, barriers, and potentials. We described key concepts of HRI such as LOA and the characteristics that define a human-robot scenario and how can we categorize and study them. As our focus is at the agroindustry sector, we saw the importance and specificity that robots could have in this industry and why applying HRI at our study is of relevance. For this case, the importance of security and the characterization of the level of integration and the nature of the relation between both team members was presented to better help us have a glimpse of the complexities that could arise on this hybrid teams.

We then moved to the plausible consequences that could arrive on having hybrid teams on organisations. One of the main impacts being how we control these teams and the shift on middle management, managerial positions, and low-skilled jobs. To further

[3] Specifically for these complexities, the authors (Ahmad, Cary, Zaheer, Ismail, & Imran, 2021) promote the idea that HRM and HRI can greatly benefit from the study of gaming literature and performance evaluations and frameworks from actual games as these scenarios of having hybrid human-machine teams that go alongside have already been in place for quite some time.

investigate specifically how will managerial sciences and HRM be affected by the rise of hybrid teams on a strategical-managerial level and already discuss the conceivable scenarios that could arise.

Machine behaviour.

To discuss this section, we can start with a powerful thought from (Rahwan, Cebrian, Obradovich, & et al, 2019) were a note from the Nobel laureate Herbert Simon on his 1969 book *Sciences of the Artificial* (Simon, 1969) was quoted :

"Natural science is knowledge about natural objects and phenomena. We ask whether there cannot also be 'artificial' science—knowledge about artificial objects and phenomena".

Here the authors gave us this insightful quote to help us better acknowledge that machine behaviour is a new interdisciplinary field of study that is emerging, as intelligent machines are "a class of actors with particular behavioural patterns and ecology" (Rahwan, Cebrian, Obradovich, & et al, 2019). Although computer science and robotics also study the same agents and it could overlap, here the focus is empirically, machine behaviour couldn't be fully understood without the whole study of the social environments where algorithms shall operate. In one phrase, is "the scientific study of behaviour exhibited by intelligent machines" (Rahwan, Cebrian, Obradovich, & et al, 2019).

The authors emphasize the importance of this field of study as they found that as Ai-powered agents are everywhere in our daily lives (see Fig.1), they will continue to expand their presence and foremost, take physical form and be among us. Moreover, they specifically recall that "when systems learn from data, their failures are linked to imperfections in the data or how data was collected" (Rahwan, Cebrian, Obradovich, & et al, 2019) calling this complex failures and results "black boxes" (Voosen, 2017). They receive an input and give an output, but the exact functioning processes and logarithms are sometimes impossible to describe and visualize in full scope (Szegedy & Et al, 2013). Moreover, the size of the data (big data) creates another layer of complexity to understand machine behaviour (Lakkaraju, Kamar, Caruana, & Horvitz, 2017).

Because of these complexities, the study of machine behaviour (and specially the nature of black box algorithms in real-world settings) should be an interdisciplinary field (see Fig.2) (Rahwan, Cebrian, Obradovich, & et al, 2019). According to the authors, this integration it's at a nascent stage, moved by the motivation on better understanding

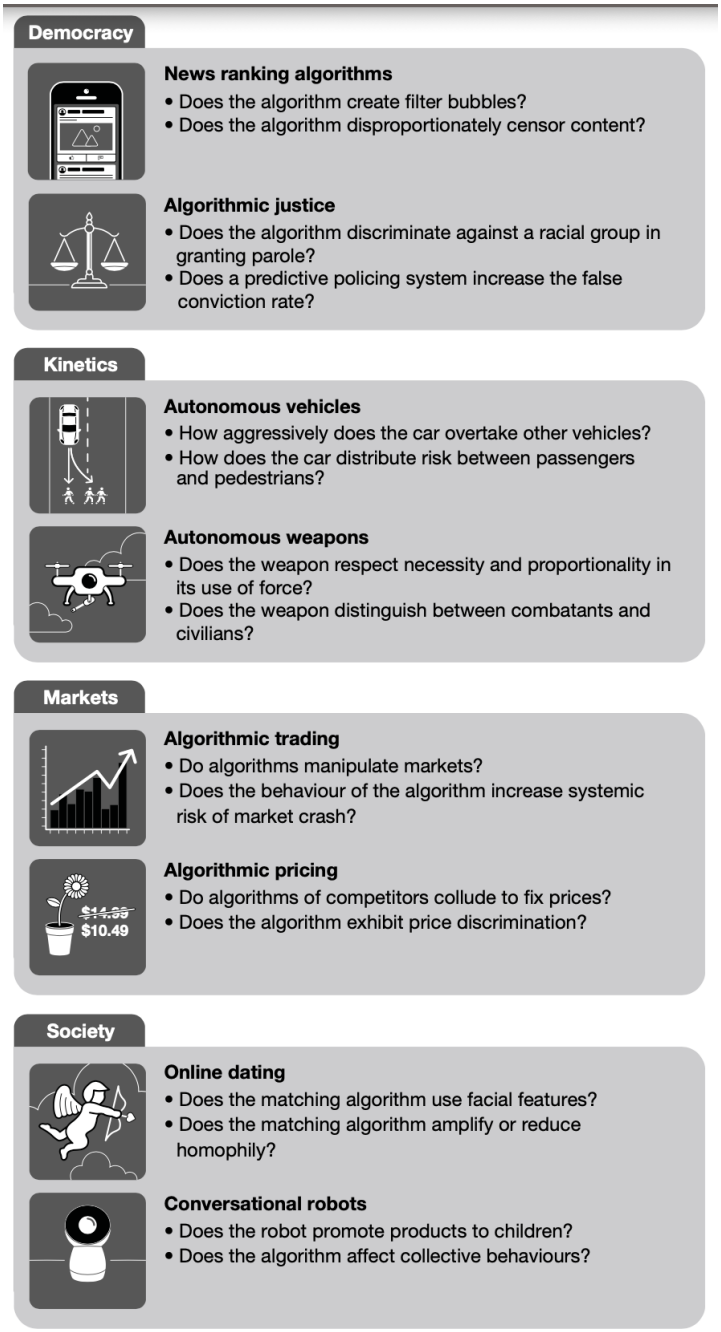


Figure 1 - Examples of questions that fall into the domain of machine behaviour.

machine behaviour. As largely the machines are studied by the same people that created them (with a mathematical science background) they are not trained on behaviourism and how to approach it correctly; and inversely, behaviourist scientist is not knowledgeable of the core algorithms found inside the machines.

Moving on how behaviourist do they research on machines, they don't focus on [4]very specific results or tasks are machines are usually accustomed to. Instead, they take in consideration the micro & macro environment where machines are, the outcomes that they produce in these (possible multiple) environments, and whether human interaction (or invasion) alters the results.

Following (Rahwan, Cebrian, Obradovich, & et al, 2019) reference, the 1973 Nobel winning scientist in medicine, Nikolaas Tinbergen, a founding member of the ethology field, proposed four dimensions when studying animal behaviour (Tinbergen, 1963). These **m, development, and evolutionary history** of a specific behaviour, providing a framework for behaviourism study (see figure 3 as a summary). In our study, we can extrapolate these dimensions following Rahwan et al, (2019) example. Machines have actions (mechanisms) that create a behaviour, they receive information from the environment that they translate into specific reactions, they action, and reactions produce consequences in their organisations and in society (making

[4] Figure 1. Taken from (Rahwan, Cebrian, Obradovich, & et al, 2019), page 479.

machines more or less common in those spaces) and exhibit evolutionary histories over past experiences (histories and human decisions) continue to influence their behaviour.

Concerning the mechanisms for generating behaviour, they depend on both the environment and the algorithm that the machine is embedded in. Moreover, more complexly, machine learning algorithms take over the information needed for their actions and reactions from huge data bases we have to consider interpretation methods of the whole system of machine learning and not only the base algorithms.

[5]

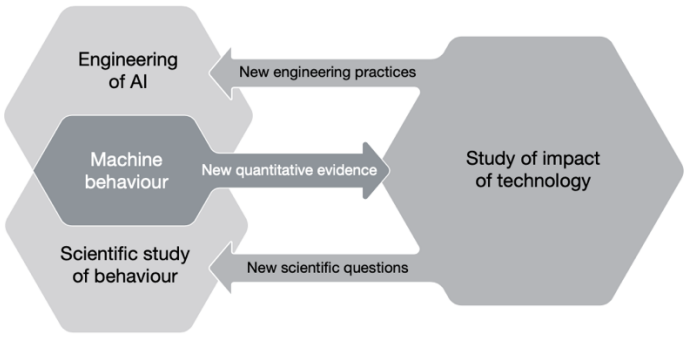


Figure 2 - The interdisciplinarity of machine behaviour

Concerning the development of behaviour, in classical animal behaviour studies the development refers on how a specific individual acquires a particular behaviour (note the distinction of long-term evolution trends). In the case of machine, this could be directly attributed to human engineering, design, and programming. The design

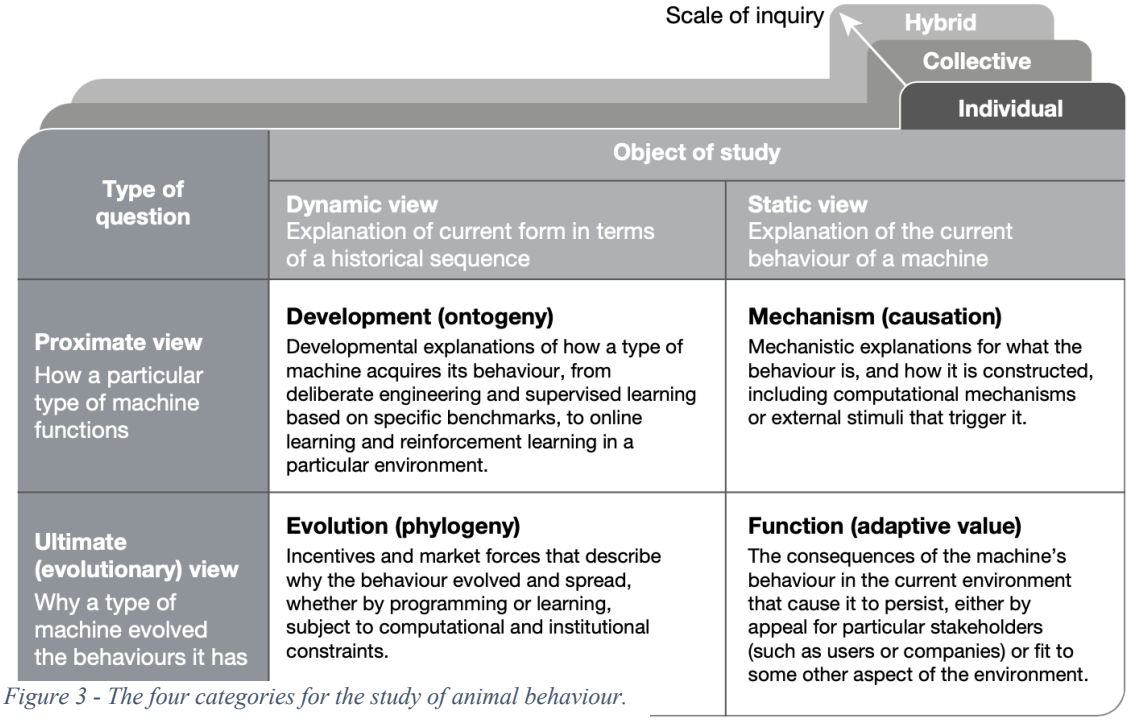


Figure 3 - The four categories for the study of animal behaviour.

[5] Figure 2. Taken from (Rahwan, Cebrian, Obradovich, & et al, 2019), page 480. *"Machine behaviour lies at the intersection of the fields that design and engineer AI systems and the fields that traditionally use scientific methods to study the behaviour of biological agents. The insights from machine behavioural studies provide quantitative evidence that can help to inform those fields that study the potential effects of technology on social and technological systems. In turn, those fields can provide useful engineering practices and scientific questions to fields that examine machine behaviours. Finally, the scientific study of behaviour helps AI scholars to make more precise statements about what AI systems can and cannot do".*

choices made by the humans will determine or highly influence the behaviours exhibited by the machine. This could be further enhanced by software & hardware upgrades, specific training scenarios that the machine was exposed to (particular datasets from where the machine performed it's tasks[6]), and finally and more complexly, their own experience (were the new concept of reinforcement learning agents appears).

Concerning function, we focus on why a specific set of behavioural mechanisms continue to exist and others don't, the importance here is to understand function as in the study of animal behaviour were the fitness (see also survival) of the agent versus the environment is crucial. In our case, successful algorithms are copied and pass over other Ai agents, the algorithms may pose incentives on how to work, these programmable incentives are of high importance as they are crucial to the machine interaction.

Concerning evolution, here the target is on how the behaviour shifted over time. In our machines study, evolution means the dependence on a specific set of technical software & hardware that at the beginning of the development of machines and Ai it was passed over the newer generations and now they are constrained by these primordial designs and dependences. However, humans can drastically shift these inherited systems (for example a software upgrade could "mutate" the machine behaviour and "spread" to other machines). Regulatory constraints could also affect the normal path of "evolution" that the machines present. As discussed, machine behaviour evolution is much complex and multi-directional that the organic evolution of animals (and humans). [7]

[6] An interesting point is if the datasets used are of quality, and the plausible biases that could arise from an Ai powered machine that was "trained" from these biased datasets.

[7] Figure 3. Taken from (Rahwan, Cebrian, Obradovich, & et al, 2019), page 481. *"The four categories Tinbergen proposed for the study of animal behaviour can be adapted to the study of machine behaviour. Tinbergen's framework proposes two types of question, how versus why, as well as two views of these questions, dynamic versus static. Each question can be examined at three scales of inquiry: individual machines, collectives of machines and hybrid human–machine systems".*

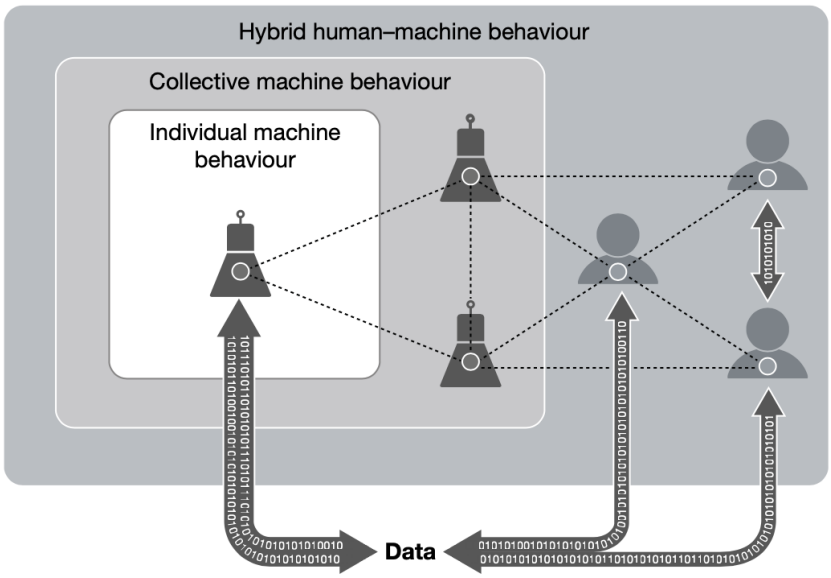


Figure 4 - Scale of inquiry in the machine behaviour ecosystem.

[8]

To finalize the theory, the authors propose a three-level scale of inquiry (see figure 4). These levels are individual machines, collectives of machines and groups of machines mixed with humans in hybrid or heterogenous systems (Rahwan, Cebrian, Obradovich, & et al, 2019). The analogy taken from the animal behaviour would be the study of a specific species, the species among their peers, and the species alongside the environment.

At the individual level, the study focuses on the intrinsic characteristics of the subject. As discussed by the authors, the centre of attention is a within-machine approach were the coding and programming used is at the centre and the interest is at how the machine reacts to changes in conditions.

At the collective level, we see the interaction and systemic-wide behaviours of the agents. Such as in the animal world, properties as higher environment awareness at a group level than at individual one, consensual decision-making, swarm-like properties could be seen. The interaction of individual agents could create (and that's the focus on) higher-order characteristics. One special concern is if machines develop social learning, where machines learn from machines, or machines learn from humans, and moreover, humans learning from the machines. As machines can (or could) instantly communicate within each other it makes an important difference from the animal behaviour and the plausible systems of interactions that could emanate from it.

[8] Figure 4. Taken from (Rahwan, Cebrian, Obradovich, & et al, 2019), page 482. *"AI systems represent the amalgamation of humans, data and algorithms. Each of these domains influences the other in both well-understood and unknown ways. Data—filtered through algorithms created by humans—influences individual and collective machine behaviour. AI systems are trained on the data, in turn influencing how humans generate data. AI systems collectively interact with and influence one another. Human interactions can be altered by the introduction of these AI systems. Studies of machine behaviour tend to occur at the individual, the collective or the hybrid human–machine scale of inquiry".*

At a hybrid human-machine level, the complex relationships between the two types of agents and how one alters the other. Here we can have three scenarios, the first one being machines shaping human behaviour. Questions of how the implications of machines and algorithms in our daily life alter our realities (consider matching algorithms by date applications, news feed choices in social media, advertising, etcetera). And beyond altering, how could they permanently change our societies and the way we live. One important point for managerial sciences could be the question on how humans react to the coproduction processes of services and goods alongside Ai and machines (Brynjolfsson & Mitchell, 2017). Other obvious scenario is humans shaping machine behaviour, as the direct creators and engineers of Ai systems it has wide implications on how machine behave. How the engineering processes done alter the machine behaviour, how the date used to train the machine also alters the behaviour or whether both engineering and data are responsible for specific actions are valid questions at this scenario. And lastly, human-machine co-behaviour, where we examine the characteristic of human-machine interactions that could include cooperation, competition and coordination behaviours and which factors facilitate (or halt) these behaviours. Questions on augmentation vs automation capabilities of machines/Ai on humans, the complexity of the feedback between the two agents, and plausible trust issues between them are of main concern also.

All the above complex theory makes an important point, studying machine behaviour is vital to ensure that we, humans, will be able to control actual and future realities, were human-machine teams will exist. To help us comprehend the underlying complexity on how the environment (and also data) can shape machines, how do they change us, how do we change them and how do we co-evolve together.

Investigation and questioning

Now that we have an overview of the main theoretical points, discussions, and complexities over Ai, Human-Robot Interaction and Machine Behaviour we can focus on some particular topics to find out if we can have a clear view on them. Firstly, coming back to the discussion over trust, a main point and always of novelty over society and the workplace as well. To have a technology widely used and accepted it must be, foremost, trusted. And as we discussed previously over human-machine teams, moreover in this situation trust is an elemental point as we are speaking about a relation that has to work and maintain over time, besides the adoption of the technology itself.

For this reason, the first hypothesis to develop will be:

- **Trust** within a human-machine team is higher when the members employ a **co-evolution approach** rather than an **individualistic one**.

What I'm wanting to investigate with this first hypothesis is which working approach managers should have in the future when working with a human-machine team. To portray the concept of co-evolution approach we can use several of the past concepts and ideas from the literature review and create a vast concept that, putting it simply, means a highly connected, dynamic, and evolving way of constructing relations and working with robots. What characterises these relations is a set of behaviours and objectives that help promote a collaboration between the two agents, an easy flow of information, a natural communication with the least barriers possible in it and mutual understanding built over time, experiences and learning of the behaviour of the other agent (evolution). We can list in simple terms and visions what describes a co-evolution approach:

a) Four aspects of work can largely be enhanced by the use of a human-machine team rather than in a human-human team : productivity, creativity, security, and commitment (to the work done).
b) Large independence of the machine from the human (imitating the liberty that the human naturally has).[9]

[9] On this point what can we understand is that robots over this approach have a high degree of liberty and capacities to independently work rather than having the human agent having a total control and supervision at all times of what the robot can and can't do and a very strict set of boundaries in their actions and communication.

c) Easy responsibility sharing in human-machine teams (over the work realized by both agents in the case of control/supervision).
d) Fluent communication and space sharing over the workplace.
e) Capacity of robots to learn from new scenarios and evolve to new conditions; having largely implanted the capacity of adaptation.
f) Sharing tasks between the two agents (humans and machines).
g) Sharing objectives and principles between humans and machines while working at the same team, space (environment) and (organizational) level.
h) Robots having a largely implanted capacity of emulating human preferences when working.

All these points envision a team that is highly connected and related to each other, open to evolve and adapt from each other and from new scenarios, fluid communication is a building block, and responsibility and tasks are equally shared.

By an individualistic approach we can envision a different scenario of human-machine teams. Here machines and robots work together but the dynamics of the relation are completely different. Both parts have their own unique set of characteristics, preferences, learning processes, communication styles, space of work and their work is much less interconnected than in the co-evolution approach. Equally as before, we can list in simple terms and visions what describes an individualistic approach:

a) Four aspects of work can be enhanced/kept at the same level by the use of a human-machine team rather than in a human-human: productivity, creativity, security, and commitment (to the work done).[10]
b) Robots are highly dependent from humans.
c) Responsibility sharing is conflictive and not clearly understandable to trace errors.
d) Communication and space sharing over the workplace are strictly limited.
e) Robots have a repetitive and predictable behaviour in their work, the human agent has to modify the robot behaviour if needed.

[10] An explanation for this term comparing it to present on co-evolution is that in both we can consider that these four factors (productivity, creativity, security, and commitment) can be enhanced in human-machines teams than in human-human teams for both approaches, co-evolution and individualistic. The difference that we can list is that in co-evolution approach these factors are largely enhanced and promoted whereas in individualistic I estimate that these factors may stay at the same level or slightly be enhanced (or some of them) comparing it to a human-human team.

f) Tasks are strictly divided and sequenced so that each agent has a set of their own.
g) Robots and human have a distinctive set of objectives and principles over their work.
h) Robots present a distinctive and unique set of behaviours and aren‘t trained to emulate human preferences.

All these points envision a team where each agent is highly independent in their work, behaviour and actions from the robot’s agent are highly dependent on human coding and establishment, we are able to predict the robot action and they don‘t present the capacity of learning or evolving from new scenarios. Communication is constricted to a set of specific scenarios and vocabulary, each part of the teams has their own set of tasks, hence, responsibility sharing is conflictive, and errors are hard to track.

Each approach of the plausible way human-machine teams could work create strictly different ways of seeing robots, Ai, and the ethics on it. In co-evolution we will tend to think that the “freer” the robots are, better they will perform, learn, and evolve (and also having these capacities) in their work. We envision a workplace where responsibility is easily shared and in the face of an error, we can track what went wrong as a whole system, rather than separating the agents as individual parts. Machines and humans would be highly interconnected in their work and the workspace will be shared. Robots will have the capacity of emulating human preferences and communication, with the objective of improving the teamwork capacity, relation, and trust (specially from the human part as we can‘t claim for the moment that robots would present trust issues). Whereas in the individualistic approach, robots present a very strict set of programmed actions, ways of communication, work environment and tasks to realize. We don‘t concretely sees robots as part of the team but more likely as augmentations of human’s capacities over a specific set of tasks that robots should concretely do. The robots, in a more ample way of seeing it, remains as a piece of machine that works and does as told, no new behaviour could be seen without humans modifying it and communication doesn‘t feel “natural” to the human part but highly “artificial”.

Before we profoundly studied trust in human-machine teams and proposed that a co-evolution approach will help for this matter rather than implementing an individualistic approach. Now moving to our second hypothesis I will focus over the

communication factor of human-machine teams. To have a good team we presented before the importance of trust, a second term proposed and completely important is communication. In any team there will be a flux of data and information being transmitted to one team member to another, this set of data-fluxes can highly be disrupted or promoted according to how efficient the communication is being done. A team will present at least a set of two members, and as in any data flux one will be the receiver and the other one the sender and in the middle is all the data being transmitted. As imagined these roles of sender-receiver will be exchanged all the time. The same will be the case for a human-machine team, hence the importance of having a good communication within the team.

For this matter I propose a second hypothesis that focuses on how we can improve these data-fluxes (communication) over human-machines teams, specifically over the agribusiness sector. I project, as before, two scenarios; but for this second one we will talk about interface & interaction experience. From one side we can have a human-like interface & interaction experience and on the other a computer-like.

The hypothesis will be as followed:

- **Communication** in human-machine teams within the agribusiness sector will be higher when using robots with a **human-like** interface and overall interaction experience rather than a **computer-like** one.

What we will focus is on the overall interaction experience & interface that human users would have over these human-machine teams and which one of them (human or computer like) better enables communication. What I mean by a human-like interface & interaction experience is a set of design premises and behaviours that enable a similarity to human relations and ways of communication. The distinction of the ways of communicating between humans and machines are actively studied to reduce any frictions or differences that machines could present when communicating. The overall experience should be in natural language, with an imitation of emotions and a sense of friendliness over the communication. Moreover, the interface present in robots should be interactive (presenting a different set of movements, sounds with the objective of imitating emotions) and familiar to the human agent to communicate with. Such as in the first hypothesis, we can list in simple terms and visions what describes a human-like interface & interaction experience:

a) Communication from the robot agent has a focus over past experiences, practises, and styles of talking present within the team members.
b) The robot agent presents a high capacity of sharing imitations of emotions and his ideas orally and physically (nonverbal communication) similar to as humans do (sense of familiarity over the conversation and movement).
c) Robots would have a high degree of responses and active feedback with their teammates.
d) Robots would present a set of states and imitated emotions to different scenarios.

All these points state an interface & interaction experience that seeks to copy the human ways of communication, whereas orally or not. Hence the amplitude seek in robots to present different imitations of emotions, large capacity of reaction and movement over human interaction and feedback. Correspondingly the potential to give the impression of familiarity in the conversation, were past experiences, practises and even styles of talking could be learned from the robot with the objective to give humans the sensation of a team affinity and belonging.

By a computer-like interface & interaction experience is a set of design premises and behaviours that create the sensation of clearly communicating to an artificial agent (robot). The focus will be over the precision of the data shared over the communication, the fastness and responsiveness of the robot to the human's commands and a language that is understandable but rather technical and pre-programmed. As before, we we can list in simple terms and visions what describes a computer-like interface & interaction experience:

a) Communication from the robot agent has a focus over precise communication (over the selection of words, grammatical construction & clear intonation).
b) The robot agents present a great capacity to listen and respond to human's commands rapidly and efficiently.
c) Robots would have precise set of movements, sounds and interactions with their teammates with the only objective of realizing their work.
d) The robot agent would present a constant look (state) and behaviours that don't change over different scenarios.

All these points state an interface & interaction experience that seeks to focus over the precision of the communication given and received from the robot rather than copying

human ways of communicating. His language and states are the same for every team member, as also his movements and any other nonverbal communication present would be predictable. Additionally, the robot would focus on having a total availability, quickness, and preciseness over his response, rather than trying to imitate a human communication were efficiency over the communication is not a daily objective that we have in our minds.

Such as before, each interface & interaction experience would create highly different robots and the vision that they present to the humans they could work with. In human-like we think of robots of members of the team that want to communicate as simply and easily as humans do, having a deep sense of familiarity and affiliation to the group where it involves his work. We can imagine a very complex robot that displays an ample set of movements and sounds, with the objective of imitating human emotions and states whether orally as also nonverbally. The focus will be in reducing communication frictions and differences between the humans and robots. Whereas in computer-like we think of robots with their own ways of "artificially" communicating. The focus is on efficiency, preciseness and total predictiveness over his behaviour. We could envision robots as complex machines that only communicate to do their work and in a technical way, but the "liberty" of movements and states is much more restricted than in the human-like.

Both hypotheses try to give an answer over which types of robots could we present in the future that enhances human trust and communication over their fellow artificial team members. As the complexities and responsibilities that we could expect at a strategic level human-machine team are enormous, hence the importance of well designing and having the right focus over which type, and which future of robots do we want and do we need.

Methodology

For the purpose of giving a response to both hypotheses, I envisioned a quantitative analysis using an online questionnaire that was transmitted to a diverse population from several countries. Next, I will present the schema of the questionnaire used. For a full overview of the questionnaire and how did it looked to the respondent's eye's, you can find it on annex 1 at the end of the writing.

Social demographics

For the purpose of data classification and plausibly for seeking relations between some social-demographical factors and the answers given, some questions have been given to respondents at first for this matter. Data tagging will be made over age, country of origin, highest level of education obtained, belonging to the agribusiness sector (working for a company directly or indirectly concerned over the food supply chain). The last tag concerning belonging to the agribusiness sector will be used as a leverage point for more specifically study if there are some overall differences in the responses given by the respondents that do belong to the agribusiness sector.

I. Age.
   1. 18-24
   2. 25-34
   3. 35-44
   4. 45-54
   5. 55-64
   6. 65-74
   7. 75+

II. Country of Origin.

III. Level of highest education obtained.
   1. Less than high school
   2. High school
   3. Technical – some college
   4. College
   5. Master
   6. PhD

IV. Do you work in the agribusiness sector ? (Every company directly or indirectly concerned over the food supply chain)

- Yes
- No

A first introductory question concerns what respondents believe to be the factor that contributes to create a good team. The focus here will be the percentage of respondents that believe that *"good communication between its members"* is the answer. As the second hypothesis concerns communication between the human-machine teams, it enables us to have a grasp of whether communication is important for the creation of a team. A second question will be specifically over robots, on whether respondents trust generally robots to perform their work well, the answers will give us a general view over what respondents think of the performance of robots and also as an introductory question to several questions concerning robots and their use in the workplace.

Trust:

1 - What creates a good team?

- Good communication between its members
- Similarity of background or personality between its members
- High level of education of its members
- Good leadership

2 - Generally, do you trust robots to perform their work well?

- 1 – 5 Not at all – Absolutely

Concerning the first hypothesis: Trust within a human-machine team is higher when the members employ a co-evolution approach rather than an individualistic one. A set of questions will be used as indicators for the likeness of respondents for one approach or the other. As explained before with the lists of terms and vision that describe each approach and each interface & interaction experience, we will use these lists of terms transformed into questions to use them as a set of indicators for one point or the other. All the questions that will continue have a scale from 1 to 5, were 1 is a negative answer over the question/scenario/phrase proposed (totally disagree) and were 5 is a positive answer over the question/scenario/phrase proposed (totally agree). Then we can study the results by prorating the answer given (transforming it into a number) and balancing the

results over one approach or the other for each specific question, and then for all of them in general. The below questions/scenarios/phrases proposed have a positive impact over the acceptance of the co-evolution approach. Altogether, a negative answer over one question will immediately prorate to a higher likeliness of the opposite approach (individualistic).

Co-evolution approach:

1. "The use of robots at the workplace helps improve productivity".
2. "The use of robots at the workplace helps improve creativity".
3. "The use of robots at the workplace helps improve security".
4. "The use of robots at the workplace helps improve commitment (to the work done)"
5. "The more independent of the human the machine are, the better I think they will perform".
6. "Sharing teamwork responsibility in a human-machine team is easier than in a human-human team."

The below questions/scenarios/phrases proposed have a positive impact over the acceptance of the individualistic approach. Altogether, a negative answer over one question will immediately prorate to a higher likeliness of the opposite approach (co-evolution).

Individualistic approach:

1. "A hybrid team (with both humans and machines as team members) is more secure when both parts are separated and don‘t communicate to each other"
2. "A hybrid team is more productive when both parts are separated and don‘t communicate to each other"
3. "A hybrid team is more creative when both parts are separated and don‘t communicate to each other"
4. "A hybrid team is more committed to work when both parts are separated and don‘t communicate to each other"
5. "I prefer to work alongside a robot that doesn‘t adapt and evolves to new conditions. Rather he should have a repetitive behaviour, even to new scenarios”.
6. "I prefer humans and machine having exclusive tasks rather than sharing them in

hybrid teams"

7. "Machines can largely excel in tasks that the human can‘t, and vice versa"
8. "Human and robots should have different objectives and principles while working at the same level and place".
9. "Robots should have their own behaviours instead of trying to imitate our preferences”.

Now moving over the second hypothesis, we also start with an introductory question on which element is more important to promote good communication within a team. The focus here will be on which response the respondents give, if there is response “a” or “c” they are much likely to be in correlation to a human-like interface & interaction experience as it correlates to the importance of familiarity and the whole sense of communication (sharing of emotions, nonverbal communication, etc) whether responses “b” or “d” focuses over efficiency and efficacity of the data-fluxes over the communication process.

Communication:

1. What is more important to enhance good communication within a team ?
    a. Past experiences and practises established within the team members.
    b. Precise use of words, good grammar, and intonation and/or good writing.
    c. Capacity of sharing emotions and thoughts orally and physically (nonverbal communication).
    d. Availability and openness to listen and respond quickly.

Concerning the second hypothesis: Communication in human-machine teams within the agribusiness sector will be higher when using robots with a human-like interface and overall interaction experience rather than a computer-like one. A set of questions will be used as indicators for the likeness of respondents for one interface & interaction experience or the other. The below questions/scenarios/phrases proposed have a positive impact over the acceptance of human-like interface and interaction experience. Altogether, a negative answer over one question will immediately prorate to a higher likeliness of the opposite interface & interaction experience (computer-like).

Human-like interface & experience:

1. “I would find it uninteresting if a robot doesn‘t shows me some kind of feedback (sound or movement) when interacting with me”.
2. “When communicating with a robot I would like to see a sense of familiarity and friendship over the talk”.
3. “If a robot displays different states and behaviours to imitate emotions, it will make him look more relatable”.

The below questions/scenarios/phrases proposed have a positive impact over the acceptance of computer-like interface and interaction experience. Altogether, a negative answer over one question will immediately prorate to a higher likeliness of the opposite interface & interaction experience (human-like).

Computer-like interface & experience:

1. “The new realism of graphics, and especially of humans, in virtual reality, worries me”.
2. “Seeing a robot that could move, act and talk, similarly to humans, worries me”.
3. “I find it reassuring when robots don‘t tries to behave like a human”.

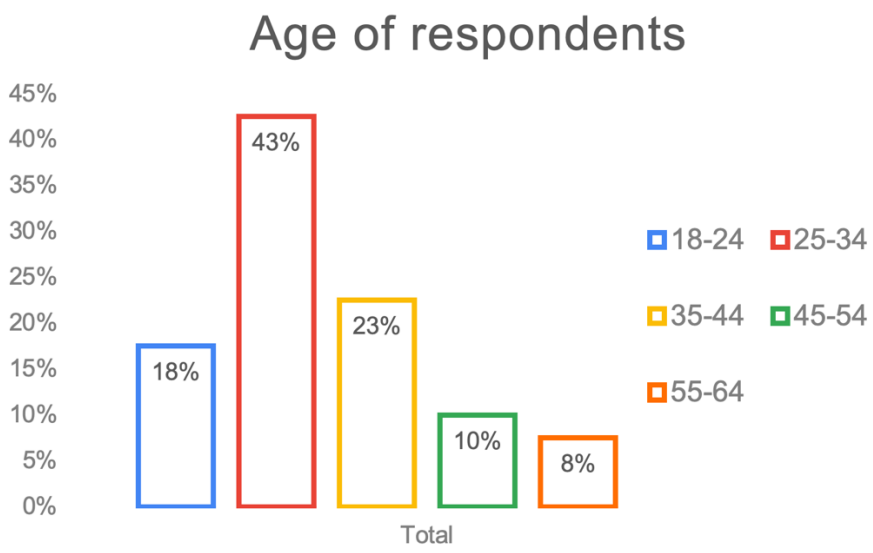
Data description

Over the following section we will explore the responses given following the questionnaire that we exposed and have a general view of the answers and visualizations that the data shows from a rapid glimpse of the answers without any analysis behind it. Then we will move on to the analysis of the data and the correlations between the hypotheses proposed, the variables and the answers given.

Social demographics

Starting with the social demographics, we can see that the majority of the respondents (61%) are between 18-34 years old, and the rest (41%) from 35-64 years old. Concerning country of origin, almost half of the respondents (45%) are Argentinians, 37% French and the rest 18% from the rest of the world. We can also dive it by continent, counting the American continent for 53%, Europe for 45% and Asia for a 2%. Concerning level of education, 50% of the respondents have a master's degree, 46% bachelors' level and below and a 4% for PhD. And lastly, a 35% of respondents work in the supply chain of the agri-business sector.

Figure 5 - Age of respondents



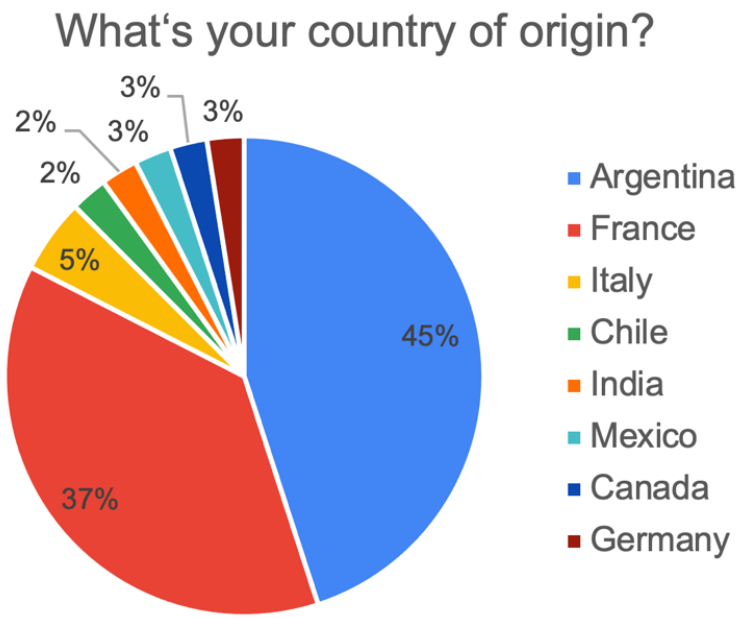


Figure 6 - Country of origin of respondents.

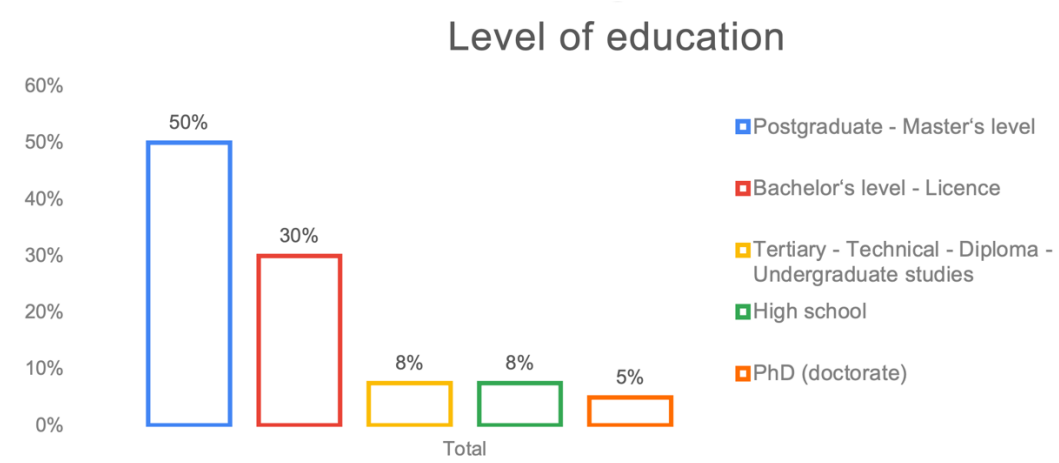


Figure 7 - Level of education of respondents.

Below are the results to the three yes/no questions proposed. Concerning on what creates a good team, 82% of respondents chose good communication between its members and the remaining 18% good leadership, no answers were given for neither "High level of education of its members" and "Similarity of social background or personality between its members". Concerning if respondents trusted robots to perform their work well, 78% partially or totally agree, 16% neither agree or disagree and the remaining 6% partially or totally disagree. And finally, on what is more important to enhance good communication within a team, 50% chose "Availability and openness to listen and respond quickly", 28% "Capacity of sharing emotions and thoughts orally and physically (nonverbal communication)", 12% "Past experiences and practises established

within the team members" and finally, the remaining 10% "Precise use of words, good grammar, and intonation and/or good writing". Meaning that 60% of respondents chose phrases functional to computer-like interface & interaction experience communication proposal and 40% chose phrases functional to human-like interface & interaction experience.

Figure 9 - Percentage of respondents working within the agribusiness sector.

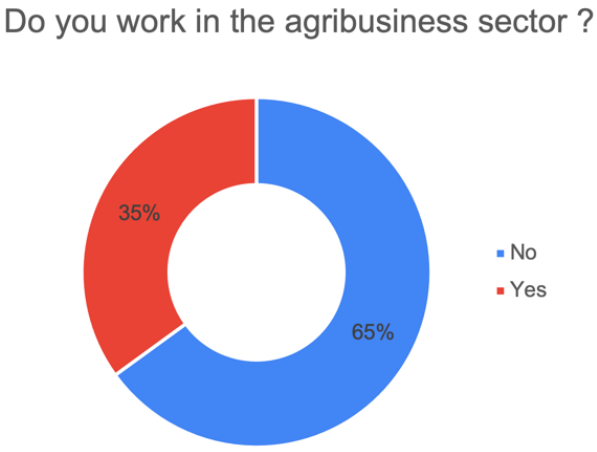


Figure 8 - Percentage of respondents who chose one of the four available answers on what creates a good team.

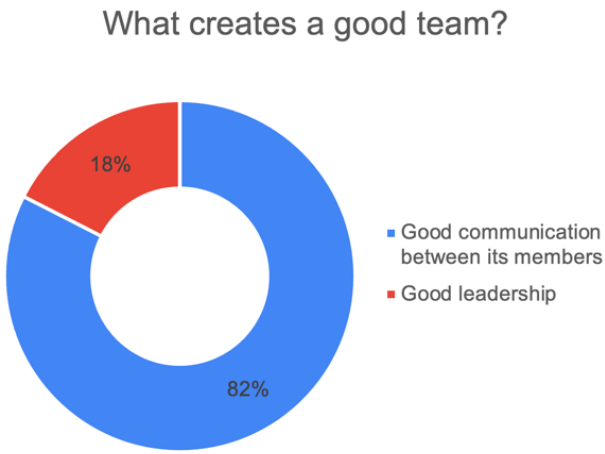


Figure 10 - Respondents trust over the performance of robots (1 as totally disagree and 5 as totally agree).

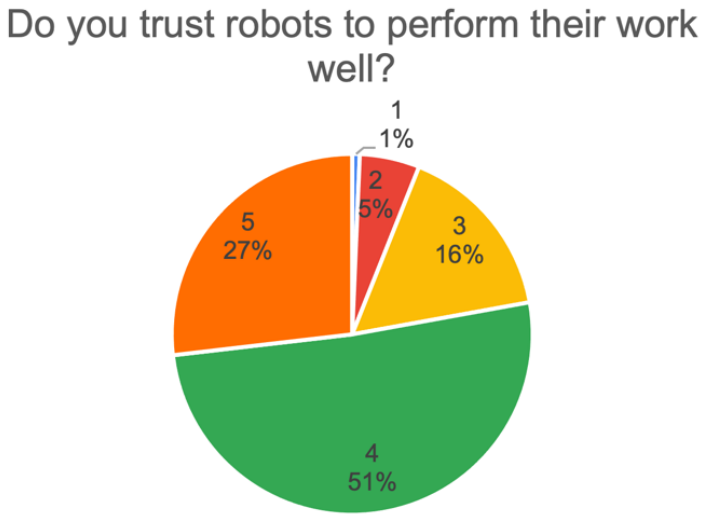
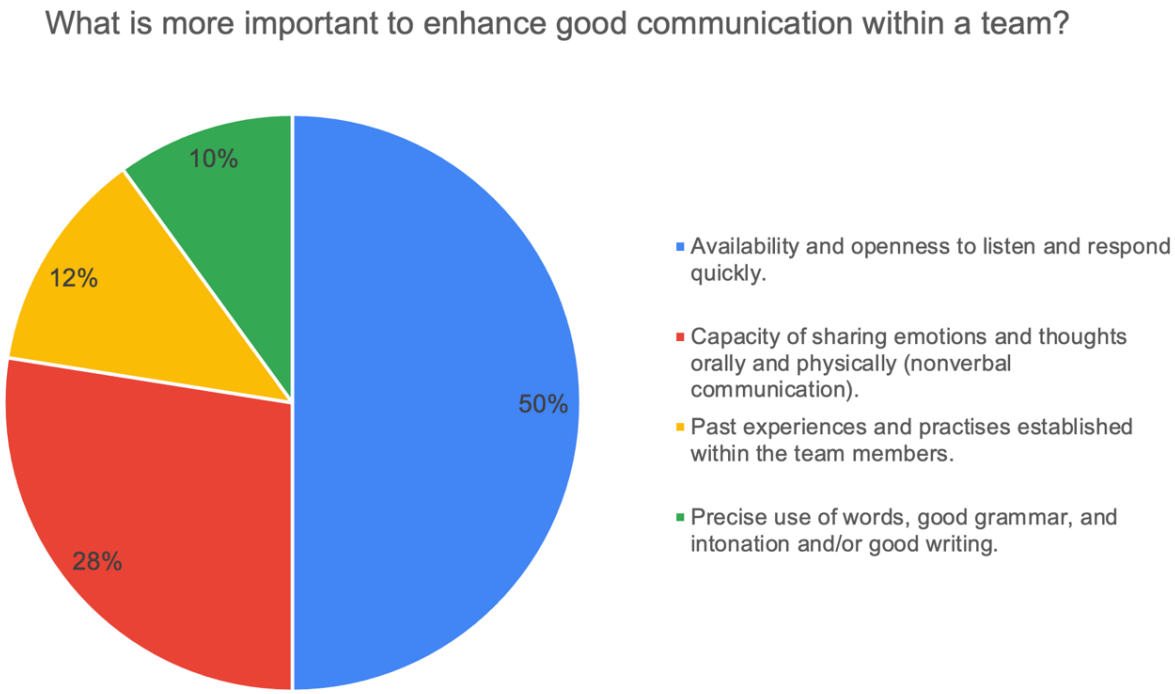


Figure 11 - Respondents choice in percentage over what is the most correct phrase on what enhances good communication.



Below there are listed a set of tables concerning each question of the questionnaire, the total value of the sum of the responses (remembering that the answers were given a scale from 1 to 5, were 1 meant totally disagree and 5 totally agree), the average value and the N percentage. We will use the N percentage as a measuring number for the analysis of the data, the closer to 1, the higher the chances that the question/phrase is accepted for the respondents. The tables are separated for hypothesis and for each element of the respective hypothesis (co-evolution & individualistic approach, and human-like and computer-like interface and interaction experience). At the end of each table there is a total average of the results for all the questions presented at the table, given a total average result for the value, average value, and N percentage value for the concerned element of the hypothesis (co-evolution & individualistic approach, and human-like and computer-like interface and interaction experience).

Table 1 - Hypothesis 1 questions

Hypothesis 1 – Questions with positive feedback for co-evolution approach.	**Value**	**Average**	**N percentage**
1. "The use of robots in the workplace helps improve productivity"	166	**4,15**	**0,83**

2. "The use of robots in the workplace helps improve creativity"	102	2,55	0,51
3. "The use of robots in the workplace helps improve security"	156	3,9	0,78
4. "The use of robots in the workplace helps improve commitment (to the work done)"	130	3,25	0,65
5. "Sharing teamwork responsibility in a human-machine team is easier than in a human-human team"	132	3,3	0,66
6. "The more independent from humans the machines are, the better I think they will perform"	120	3	0,6
Total average	**134,3**	**3,4**	**0,7**

Table 2 - Hypothesis 1 second approach questions

Hypothesis 1 – Questions with positive feedback for individualistic approach.	Value	Average	N percentage
1. "I prefer to work alongside a robot that doesn‘t adapt or evolve to new conditions. Rather he should have a repetitive behaviour, even to new scenarios"	108	2,7	0,54
2. "I prefer humans and machines having exclusive tasks rather than sharing them in hybrid teams"	112	2,8	0,56
3. "Machines can largely excel in tasks that the humans can‘t, and vice versa"	165	4,125	0,825
4. "Humans and robots should have different objectives and principles while working at the same level and place"	133	3,325	0,665
5. "Robots should have their own behaviours instead of trying to imitate our preferences"	112	2,8	0,56
6. "A hybrid team (with both humans and machines as team members) is more secure when both parts are separated and don‘t relate to each other"	108	2,7	0,54
7. "A hybrid team is more committed to work when both parts are separated and don‘t relate to each other"	90	2,25	0,45
8. "A hybrid team is more productive when both parts are separated and don‘t relate to each other"	91	2,275	0,455
9. "A hybrid team is more creative when both parts are separated and don‘t relate to each other"	104	2,6	0,52

Total average	**113,7**	**2,8**	**0,6**

Table 3 - Hypothesis 2 questions.

Hypothesis 2 – Questions with positive feedback to human-like interface & interaction experience.	Value	Average	N percentage
1. **"I would find it uninteresting if a robot doesn't shows me some kind of sensorial feedback (sound or movement) when interacting with me"**	112	**2,8**	**0,56**
2. **"When communicating with a robot I would like to see a sense of familiarity and friendship over the talk"**	114	**2,85**	**0,57**
3. **"If a robot displays different states and behaviours to imitate emotions, it will make him look more relatable"**	114	**2,85**	**0,57**
Total average	**113,3**	**2,8**	**0,6**

Table 4 - Hypothesis 2 second position questions.

Hypothesis 2 – Questions with positive feedback to computer-like interface & interaction experience.	Value	Average	N percentage
1. **"The new realism of graphics, and especially of humans, in virtual reality, worries me"**	125	**3,125**	**0,625**
2. **"Seeing a robot that could move, act and talk, similarly to humans, worries me"**	132	**3,3**	**0,66**
3. **"I find it reassuring when robots don't tries to behave like a human"**	146	**3,65**	**0,73**
Total average	**134,3**	**3,4**	**0,7**

Data analysis

Now we will discuss the deeper results and conclusions that I extracted from the data, having both the hypotheses as guiding factors to study the veracity of them both. Remembering the first hypothesis "Trust within a human-machine team is higher when the members employ a co-evolution approach rather than an individualistic one", we have trust as the central value, and two possible approaches that lead to a higher level of it within human-machine teams, co-evolution, and individualistic approach. We already discussed the questions that gave particular insights to each approach, so now we can analyse question by question (an indicator by indicator) the results.

From the set of 15 questions concerning the first hypothesis, 6 of them are positive over the co-evolution approach (or also we can say against the individualistic approach) and 9 of them positive to the individualistic approach. Meaning that as we had a scale from 1 to 5 where 1 was totally disagree and 5 totally agree, the higher the results in numbers from each concerning question (whether positive to co-evolution or individualistic approach), the higher the accordance from respondents to one of the particular factors of the approach (ex: from the question "The use of robots in the workplace helps improve creativity", the average result from the 1 to 5 scale gave 2,55, meaning that the total dispersion from this question from the median answer that is 3 gave a -23%; as the key indicator here is creativity and this question is positive to the co-evolution approach it gives and actual negative feedback to the first hypothesis). The questions and their respective indicators are as follows:

Table 5 - Hypothesis 1 - Co-evolution approach

Hypothesis 1 – Questions with positive feedback for co-evolution approach.	**Indicator**	**Percentage of dispersion from the median value (3)**
"The use of robots in the workplace helps improve productivity"	Productivity	**58%**
"The use of robots in the workplace helps improve creativity"	Creativity	**-23%**
"The use of robots in the workplace helps improve security"	Security	**45%**
"The use of robots in the workplace helps improve commitment (to the work done)"	Commitment	**13%**
"Sharing teamwork responsibility in a human-machine team is easier than in a human-human team"	Teamwork responsibility	**15%**
"The more independent from humans the machines are, the better I think they will perform"	Independence	**0%**

Table 6 - Hypothesis 1 - Individualistic approach

Hypothesis 1 – Questions with positive feedback for individualistic approach.	**Indicator**	**Percentage of dispersion from the median value (3)**
"I prefer to work alongside a robot that doesn‘t adapt or evolve to new conditions. Rather he should have a repetitive behaviour, even to new scenarios"	Adaptation / evolution	**-15%**
"I prefer humans and machines having exclusive tasks rather than sharing them in hybrid teams"	Sharing tasks	**-10%**
"Machines can largely excel in tasks that the humans can‘t, and vice versa"	Tasks differentiation	**56%**
"Humans and robots should have different objectives and principles while working at the same level and place"	Objectives and principles	**16%**
"Robots should have their own behaviours instead of trying to imitate our preferences"	Behaviours and preferences	**-10%**
"A hybrid team (with both humans and machines as team members) is more secure when both parts are separated and don‘t relate to each other"	Security	**-15%**
"A hybrid team is more committed to work when both parts are separated and don‘t relate to each other"	Commitment	**-38%**
"A hybrid team is more productive when both parts are separated and don‘t relate to each other"	Productivity	**-36%**
"A hybrid team is more creative when both parts are separated and don‘t relate to each other"	Creativity	**-20%**

The avid readers might have found that creativity, productivity, commitment, and security are repeated, for the purpose of having a more secure result over these four dimensions; the results from one question to the other have been compensated in the total results as follows.

We can summarize the total percentages of dispersion from the hypothesis 1 with a positive 12% over the median value (3 in the scale from 1 to 5) from all the responses that promotes the use of a co-evolution approach.

More specifically we can create two equations that mathematically could help to see more clearly which indicators have more weight over one approach or the other. Also, the following equation could be used for a future market study, replacing the indicators value with one that the investigator thinks eligible and then multiplying it by the percentage of dispersion (ex: the relevance of S – Security over the investigator study is

high so he can chose from a set from 1 to 5 of importance to use 5, then multiplying it by the S coefficient of 0,30 he obtains 1,5; the sum of all the ponderations of the indicators with the coefficients divided by the quantity of indicators used could give a clear number over if one or the other approach is feasible taking in consideration which indicators are more important for the investigator).

- Equation of indicators that promote the use of a co-evolution approach to gain trust in human-machine teams :

Equation 1

$$P(0,47) + S(0,3) + CC(0,25) + TR(0,15) + AE(0,15) + ST(0,10) + BP(0,10) / 7$$

I. P = Productivity
II. S = Security
III. CC = Commitment
IV. TR = Teamwork responsibility
V. AE = Adaptation/evolution
VI. ST = Sharing tasks
VII. BP = Behaviours and preferences

- Equation of indicators that discourage the use of a co-evolution approach to gain trust in human-machine teams:

Equation 2

$$C(0,01) + TD(0,56) + OP(0,16) / 3$$

I. C = Creativity
II. TD = Tasks differentiation
III. OP = Objectives and principles

Again, for the avid reader, the indicator independence has not been used, as the result gave a perfect 0% deviation from the median result (3 from the 1 to 5 scale). Meaning to our study and methodological system that independence of the robot within a human-machine team couldn‘t be used as this indicator neither promotes one approach or the other.

Moving now to our second hypothesis "Communication in human-machine teams within the agribusiness sector will be higher when using robots with a human-like interface and overall interaction experience rather than a computer-like one", we have communication as the central value and two possible paths that lead a higher level of communication within human-machine teams, human-like and computer-like interface & interaction experience. We already discussed the questions that gave particular insights to each position , so now we can analyse question by question (an indicator by indicator) the results.

From the set of 6 questions concerning the first hypothesis, 3 of them are positive over the human-like psotion (or also we can say against the computer-like position) and 3 of them positive to the computer-like position. The questions and their respective indicators are as follows:

Table 7 - Hypothesis 2 - Human-like.

Hypothesis 2 – Questions with positive feedback to human-like interface & interaction experience.	**Indicator**	**Percentage of dispersion from the median value (3)**
"I would find it uninteresting if a robot doesn't shows me some kind of sensorial feedback (sound or movement) when interacting with me"	Sensorial feedback	**-10%**
"When communicating with a robot I would like to see a sense of familiarity and friendship over the talk"	Familiarity/friendship	**-8%**
"If a robot displays different states and behaviours to imitate emotions, it will make him look more relatable"	Emotions imitation	**-8%**

Table 8 - Hypothesis 2 - Computer-like.

Hypothesis 2 – Questions with positive feedback to computer-like interface & interaction experience.	**Indicator**	**Percentage of dispersion from the median value (3)**
"The new realism of graphics, and especially of humans, in virtual reality, worries me"	Realism of graphics	**6%**
"Seeing a robot that could move, act and talk, similarly to humans, worries me"	Actions similarity	**15%**
"I find it reassuring when robots don't tries to behave like a human"	Behaviours similarity	**33%**

We can summarize the total percentages of dispersion from the hypothesis 2 with a negative 13% over the median value (3 in the scale from 1 to 5) from all the responses

that promotes the use of a human-like interface & interaction experience. Or analogically, a positive 13% over the median value from all the responses that promotes the use of a computer-like interface & interaction experience.

As with hypothesis 1, we can have an equation that mathematically could help to see more clearly which indicators have more weight over one approach or the other (for this case, all indicators weight favourably to the computer-like position, meaning that we will only have one equation).

- Equation of aspects that promote the use of a computer-like interface & interaction experience to improve communication in human-machine teams:

Equation 3

$$SF(0,10) + FF(0,08) + EI(0,08) + RG(0,06) + AS(0,15) + BS(0,33) / 6$$

I. SF = Sensorial feedback
II. FF = Familiarity/Friendship
III. EI = Emotions imitation
IV. RG = Realism of graphics
V. AS = Actions similarity
VI. BS = Behaviours similarity

Now, again, the avid reader might ask himself what about the focus over the agri-business sector as I haven‘t introduced any results or remarks over these points over the data analysis. Luckily, to finalize the data analysis we can highlight some important variations from the group that responded “Yes” to the classificatory question of : Do you work in the agribusiness sector ? (Every company directly or indirectly concerned over the food supply chain).

Concerning hypothesis 1, the result concerning the overall median value shifted, from a positive 12% of the total respondents to a lower 6% over the median value from the specific respondents that work within the agribusiness sector regarding the use of the co-evolution approach to create higher trust within human-machine teams. Although the positivity of the co-evolution approach remains, it was halved. Interestingly, again the independence indicator has a 0% value, and also the behaviours and preferences indicator.

For this reason, these indicators will be out the equations; the final look of the equations will be:

- Equation of indicators that promote the use of a co-evolution approach to gain trust in human-machine teams :

Equation 4

$$P(0,17) + S(0,14) + CC(0,08) + TR(0,16) + AE(0,08) + ST(0,12) / 5$$

- Equation of indicators that discourage the use of a co-evolution approach to gain trust in human-machine teams:

Equation 5

$$C(0,02) + TD(0,22) + OP(0,07) / 3$$

Over hypothesis 2, the result concerning the overall median value shifted as well, from a negative 13% of the total respondents to a negative 4% over the median value from the specific respondents that work within the agribusiness sector regarding the use of a human-like interface & interaction experience position to promote communication trust within human-machine teams. Or more simply, a positive 4% concerning the use of a computer-like interface & interaction experience. Additionally, the EI (emotions imitation) indicator becomes positive to a 2% (as also all the other indicators become near zero as well). Meaning that for the agri-business sector responses, the equations shift as followed :

- Equation of aspects that promote the use of a human-like interface & interaction experience to improve communication in human-machine teams within the agribusiness sector:

Equation 6

$$Ei\ (0,02)$$

- Equation of aspects that promote the use of a computer-like interface & interaction experience to improve communication in human-machine teams within the agribusiness sector:

Equation 7

$$SF(0,02) + FF(0,04) + RG(0,03) + AS(0,5) + BS(0,1) / 5$$

In overall, respondents from the agribusiness sector are less radical over their responses, having a lower general dispersion from the median values (compared to the total of responses).

Conclusion

Over this study we have had a global insight of the importance and always actual significance of Ai in our days. As an evolving field, Ai remains full of mysteries and situations to be discovered and discussed, in our case more profoundly about Human Robot Interaction and Machine Behaviour; with a focus on human-machine teams and more specifically, over the agribusiness sector.

Review

Remembering what the first research question was: What plausible constraints could arise within the agribusiness sector in the near future that would make managing a human-machine team different from a regular human-human team ? Nevertheless, other questions were of interest as which elements of a human-machine team are different from a human-human team, why is it of importance to study these differences, how can we approach these plausible differences, and which managerial approach shall be future leaders have within their organizations to lead them in a new reality of having hybrid human-machines teams at a strategic level. This study could help moving forward and giving some insights over this matter in the form of two specific hypotheses:

3. *Trust within a human-machine team is higher when the members employ a co-evolution approach rather than an individualistic one.*
4. *Communication in human-machine teams within the agribusiness sector will be higher when using robots with a human-like interface and overall interaction experience rather than a computer-like one.*

Report

For this matter, a questionnaire was used as a data-base creator to gain clear insights and a set of indicators with the objective of measuring both hypotheses. The questionnaire was answered by 40 respondents, from 8 countries, from different age groups and different levels of education. With a 35% of them working within the agribusiness sector, giving a base of analysis to highlight the differences between the total amount of responses.

As discussed, the questionnaire proposed had a set of phrases and questions that were used as indicators for the likeliness of one approach or the other (co-evolution or individualistic) for the first hypothesis and interface & interaction experience (human-like or computer-like) for the second hypothesis. These indicators were then analysed as

a total average deviation in percentage over the median value (3 from a scale from 1 to 5 where 1 was totally disagree and 5 totally agree). The percentages of the indicators were balanced in total giving a:

1. Positive 12% over the median value from all the responses that promotes the use of a co-evolution approach.
2. Positive 6% over the median value from the agribusiness sector responses that promote the use of a co-evolution approach.
3. Negative 13% over the median from all the responses that promotes the use of a human-like interface & interaction experience.
4. Negative 4% over the median from the agribusiness sector responses that promote the use of a human-like interface & interaction experience.

Another result was the possibility of using the following equations within the agribusiness sector as a market study over organizations. As a tool to balance-out the weight for each indicator and to gain a deeper insight on which approach or interface & interaction experience is more appropriate for the situation:

- Equation of indicators that promote the use of a co-evolution approach to gain trust in human-machine teams :

$$P(0,17) + S(0,14) + CC(0,08) + TR(0,16) + AE(0,08) + ST(0,12) / 5$$

- Equation of indicators that discourage the use of a co-evolution approach to gain trust in human-machine teams:

$$C(0,02) + TD(0,22) + OP(0,07) / 3$$

- Equation of aspects that promote the use of a human-like interface & interaction experience to improve communication in human-machine teams:

$$Ei\ (0,02)$$

- Equation of aspects that promote the use of a computer-like interface & interaction experience to improve communication in human-machine teams:

$$SF(0,02) + FF(0,04) + RG(0,03) + AS(0,5) + BS(0,1) / 5$$

Another particularity from the responses received from the agribusiness sector denote a tendency to be more moderate and cautious over the development of this technologies, as the results were much more closes to the median value (3) than in the general pool of answers. Responding to the approaching of both percentages from the hypotheses nearer to zero, making it harder to identify the highly likeliness of one

approach / interface & interaction experience or the other particularly for this sector. Not forgetting that we discussed about trust and communication, two core values present in any type of team, the study of them helps give an insight over the intricacies of hybrid human-machine teams that we could face withing 3-5 years as I envision from today's time. Finally, concerning the results from the agribusiness answer pull, it seems that this sector is much more sensible to a human-like interface & interaction experience when trying to improve communication. Although the results in general remain negative, the sharply shift from the numbers demonstrates an openness and curiosity over the similar ways of communication between humans and machines over this sector than over the general answer pull. And on the contrary, this sector also seems more insecure concerning the sharing of spaces, tasks, and objectives with machines as the lower positiveness from the hypothesis one shows. Altogether, a co-evolution approach seems more practical to promote trust within this sector and moreover the general pull.

Surely this investigation could be catalogued as exploratory, as we are trying to understand a reality that is quite far away today and from today‘s perspective of things; nonetheless, a reality that doesn‘t seems impossible either.

Limitations

Many could be the barriers faced within the borders of this investigation, but I will make out the most visible ones. Starting with the size of the population from where I studied the responses, some biases could be present as the age, education level, geographics and sector of work aren‘t that diverse enough to clearly represent a fairly demonstration of society as a whole.

Secondly, as a highly exploratory survey and study, the answers and opinions received might be far away from the reality as most of the questions are open for the respondent's imagination of how the future could be rather than how do they feel the reality is or how do they see it. Some important momentum biases could also have taken place as this questionnaire was sent some days after the global emergence of the powerful generative Ai software called ChatGPT. Over the last months several times Ai and robots have taken the news screen and the focus has been largely over the negative side of it and the speed of the changes that the society is seeing and how this could affect future generations. For this reason, I estimate quite conservatory responses over the questionnaire proposed as the respondents could have been highly agitated by the emergence of these last technologies.

Thirdly, the length and preciseness of the questions might come under discussion, as it is quite complex to summarize such vast concepts of trust and communication over less than 30 questions and moreover, to have a solid set of indicators from each value that will lead us to one result or the other. A bigger set of questions and indicators could have been proposed to achieve a more rigid methodology of study.

And finally, as the research topic concerns a vast variety of disciplines, is hard to correct set of theory that would correctly apply to the study of Ai and robotics at a managerial level within organizations. Rather, a more holistic approach was used taking ideas from HRI and Machine behaviour, mixed with human resources management to try to end up with simple, yet specific concepts and indicators to the study. Nevertheless, more solid concepts and study methods or experiments are possible to be made.

The future of hybrid human-machine teams

The results set a path to brave enough future investigators to keep studying the plausible constraints and characteristics of future hybrid human-machines teams at a strategic level, and more specifically at the agribusiness sector. One interesting approach would be to start using real life experiment to test the solidity of the indicators proposed and the equations and their percentages. It could be of much help to have a better constructed model of over which indicators encourage one approach or interface & interaction experience or the other.

I would personally find exciting to keep improving the concept of co-evolution approach as trust enhancing path rather than individualistic as it creates a totally different vision of how future hybrid machine-humans' teams could be, specially concerning ethics, communication and "liberty" of the artificial agents involved. Getting closer to prove that having artificial agents that have more "liberty" and communicate in a more "human" way is more constructive to the future development of teams, hence organizations, hence society could be quite a scientific breakthrough.

I leave the writing for curious future investigators to keep exploring the scenarios of this "not so far away" reality that might come to the workplace in a couple years from now. Maybe not so refined as we might expect but slowly and firmly Ai and robotics have proven their value and disruptive potential to humanity and organizations. Getting ahead of these changes will create only but benefit to the future societies as we will become more ready and have a slightly clear insight on what is yet to come to the workplace environment, and the future of the agribusiness sector alike.

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Annex 1 – Questionnaire

1. What‘s your age?
2. What‘s your country of origin? (Name of the country in English).
3. What‘s your highest level of education?
    a. PhD (doctorate)
    b. Postgraduate - Master‘s level
    c. Bachelor‘s level – Licence
    d. Tertiary - Technical - Diploma - Undergraduate studies
    e. High school
    f. Lower than high school
4. Do you work in the agribusiness sector ? (Every company directly or indirectly concerned over the food supply chain)
5. What creates a good team?
    a. Good communication between its members
    b. Similarity of background or personality between its members
    c. High level of education of its members
    d. Good leadership
6. Generally, do you trust robots to perform their work well?
7. "The use of robots in the workplace helps improve productivity"
8. "The use of robots in the workplace helps improve creativity"
9. "The use of robots in the workplace helps improve security"
10. "The use of robots in the workplace helps improve commitment (to the work done)"
11. "Sharing teamwork responsibility in a human-machine team is easier than in a human-human team"
12. "The more independent from humans the machines are, the better I think they will perform"
13. "I prefer to work alongside a robot that doesn‘t adapt or evolve to new conditions. Rather he should have a repetitive behaviour, even to new scenarios"
14. "I prefer humans and machines having exclusive tasks rather than sharing them in hybrid teams"
15. "Machines can largely excel in tasks that the humans can‘t, and vice versa"

16. "Humans and robots should have different objectives and principles while working at the same level and place"
17. "Robots should have their own behaviours instead of trying to imitate our preferences"
18. "A hybrid team (with both humans and machines as team members) is more secure when both parts are separated and don‘t relate to each other"
19. "A hybrid team is more committed to work when both parts are separated and don‘t relate to each other"
20. "A hybrid team is more productive when both parts are separated and don‘t relate to each other"
21. "A hybrid team is more creative when both parts are separated and don‘t relate to each other"
22. What is more important to enhance good communication within a team?
    a. Availability and openness to listen and respond quickly.
    b. Precise use of words, good grammar, and intonation and/or good writing.
    c. Capacity of sharing emotions and thoughts orally and physically (nonverbal communication).
    d. Past experiences and practises established within the team members.
23. “I would find it uninteresting if a robot doesn‘t show me some kind of sensorial feedback (sound or movement) when interacting with me”
24. “When communicating with a robot I would like to see a sense of familiarity and friendship over the talk”
25. “If a robot displays different states and behaviours to imitate emotions, it will make him look more relatable”
26. “The new realism of graphics, and especially of humans, in virtual reality, worries me"
27. “Seeing a robot that could move, act and talk, similarly to humans, worries me”
28. “I find it reassuring when robots don‘t tries to behave like a human”